# Chapter 1. The Algebra and Topology of $\mathbb{R}^n$

Linear algebra, matrix inversion and determinants, metric topology, compactness and connectedness in $\mathbb{R}^n$

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## Overview

Linear algebra and topology of $\mathbb{R}^n$. §1, §3 are review; §2, §4 may be new (elementary matrices, axiomatic determinant, $\epsilon$-neighborhood theorem, compactness of rectangles). Proofs of results Munkres leaves unproved are marked [Supplement].

# §1. Review of Linear Algebra

**Motivation.** Multivariable calculus is linear algebra applied locally: derivatives are linear maps, volume changes by determinants, forms are alternating multilinear maps.

### 1.1 Vector spaces

**Definition 1.1 (vector space)**

A set $V$ with an operation $\mathbf{x}+\mathbf{y}$ (vector addition) and an operation $c\mathbf{x}$ (scalar multiplication by $c \in \mathbb{R}$) is a **vector space** (or **linear space**) if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $c, d \in \mathbb{R}$:

(1) $\mathbf{x}+\mathbf{y} = \mathbf{y}+\mathbf{x}$;

(2) $\mathbf{x}+(\mathbf{y}+\mathbf{z}) = (\mathbf{x}+\mathbf{y})+\mathbf{z}$;

(3) there is a unique vector $\mathbf{0}$ with $\mathbf{x}+\mathbf{0} = \mathbf{x}$ for all $\mathbf{x}$;

(4) $\mathbf{x}+(-1)\mathbf{x} = \mathbf{0}$;

(5) $1\mathbf{x} = \mathbf{x}$;

(6) $c(d\mathbf{x}) = (cd)\mathbf{x}$;

(7) $(c+d)\mathbf{x} = c\mathbf{x}+d\mathbf{x}$;

(8) $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x}+c\mathbf{y}$.

The basic example is $\mathbb{R}^n$, the set of $n$-tuples $\mathbf{x} = (x_1, \dots, x_n)$ with componentwise operations $\mathbf{x}+\mathbf{y} = (x_1+y_1, \dots, x_n+y_n)$, $c\mathbf{x} = (cx_1, \dots, cx_n)$.

A subset $W \subset V$ is a **linear subspace** if $\mathbf{x}+\mathbf{y} \in W$ and $c\mathbf{x} \in W$ whenever $\mathbf{x}, \mathbf{y} \in W$, $c \in \mathbb{R}$; then $W$ is a vector space in its own right.

**Definition 1.2 (span, independence, basis, dimension)**

Let $\mathbf{a}_1, \dots, \mathbf{a}_m \in V$.

- They **span** $V$ if every $\mathbf{x} \in V$ can be written in *at least* one way as a **linear combination** $\mathbf{x} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m$.
- They are **independent** if every $\mathbf{x} \in V$ can be written in *at most* one way as such a combination; equivalently, $d_1\mathbf{a}_1 + \cdots + d_m\mathbf{a}_m = \mathbf{0}$ forces $d_1 = \cdots = d_m = 0$.
- They form a **basis** if they both span $V$ and are independent.

If $V$ has a basis of $m$ vectors, $m$ is the **dimension** of $V$ (by Theorem 1.1 this is well defined); the zero space has dimension 0.

**Theorem 1.1**

Suppose $V$ has a basis consisting of $m$ vectors. Then any set of vectors that spans $V$ has at least $m$ vectors, and any set of vectors of $V$ that is independent has at most $m$ vectors. In particular, any basis for $V$ has exactly $m$ vectors.

**Proof [Supplement].** It suffices to show: if $\mathbf{w}_1, \dots, \mathbf{w}_k$ span $V$ and $\mathbf{v}_1, \dots, \mathbf{v}_l$ are independent, then $l \le k$ (apply it with the basis as the spanning set, and with the basis as the independent set). Write $\mathbf{v}_j = \sum_{i=1}^{k} a_{ij}\mathbf{w}_i$. If $l > k$, the homogeneous linear system $\sum_{j=1}^{l} a_{ij}x_j = 0$ $(i = 1, \dots, k)$ has more unknowns than equations, hence (by Gauss–Jordan reduction: an echelon form has at most $k$ corner entries, so some unknown is free) a nonzero solution $(x_1, \dots, x_l)$. Then $\sum_j x_j\mathbf{v}_j = \sum_i(\sum_j a_{ij}x_j)\mathbf{w}_i = \mathbf{0}$, contradicting independence. ∎

$\mathbb{R}^n$ has dimension $n$: the **standard basis** is $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)$, $\dots$, $\mathbf{e}_n = (0, \dots, 0, 1)$. Every basis of $\mathbb{R}^n$ has exactly $n$ vectors.

**Theorem 1.2**

Let $V$ be a vector space of dimension $m$. If $W$ is a linear subspace of $V$ different from $V$, then $W$ has dimension less than $m$. Furthermore, any basis $\mathbf{a}_1, \dots, \mathbf{a}_k$ for $W$ may be extended to a basis $\mathbf{a}_1, \dots, \mathbf{a}_k, \mathbf{a}_{k+1}, \dots, \mathbf{a}_m$ for $V$.

**Proof [Supplement].** First, $W$ has a finite basis: choose independent vectors of $W$ one at a time as long as possible; by Theorem 1.1 the process stops after at most $m$ steps, and a maximal independent subset of $W$ spans $W$ (if $\mathbf{x} \in W$ were not a combination of them, adjoining $\mathbf{x}$ would keep independence). Now let $\mathbf{a}_1, \dots, \mathbf{a}_k$ be a basis of $W$. If they do not span $V$, pick $\mathbf{a}_{k+1} \in V$ not in their span; then $\mathbf{a}_1, \dots, \mathbf{a}_{k+1}$ are independent (a relation with nonzero coefficient of $\mathbf{a}_{k+1}$ would put $\mathbf{a}_{k+1}$ in the span; one with zero coefficient contradicts independence of the $\mathbf{a}_i$). Repeat; by Theorem 1.1 the process stops when we have an independent spanning set of $m$ vectors. If $W \neq V$, at least one vector was added, so $k < m$. ∎

## 1.2 Inner products and norms

**Definition 1.3 (inner product)**

An **inner product** on $V$ assigns to $\mathbf{x}, \mathbf{y} \in V$ a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ such that for all $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and scalars $c$:
(1) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
(2) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$;
(3) $\langle c\mathbf{x}, \mathbf{y} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, c\mathbf{y} \rangle$;
(4) $\langle \mathbf{x}, \mathbf{x} \rangle > 0$ if $\mathbf{x} \neq \mathbf{0}$. $V$ together with an inner product is an **inner product space**.

On $\mathbb{R}^n$ we use the **dot product** $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + \cdots + x_ny_n$ (not $\mathbf{x} \cdot \mathbf{y}$, reserved for matrix products).

**Definition 1.4 (norm)**

The **length** (or norm) in an inner product space is $\|\mathbf{x}\| = \langle \mathbf{x}, \mathbf{x} \rangle^{1/2}$. It satisfies (1) $\|\mathbf{x}\| > 0$ if $\mathbf{x} \neq \mathbf{0}$;
(2) $\|c\mathbf{x}\| = |c|\,\|\mathbf{x}\|$;
(3) $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ (**triangle inequality**); equivalently (3′) $\|\mathbf{x} - \mathbf{y}\| \ge \|\mathbf{x}\| - \|\mathbf{y}\|$. Any function $V \to \mathbb{R}$ with properties (1)–(3) is called a **norm** on $V$.

**Proof of (3) [Supplement].** *Cauchy–Schwarz*: $|\langle \mathbf{x}, \mathbf{y} \rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$. This is trivial if $\mathbf{x}$ or $\mathbf{y}$ is $\mathbf{0}$; otherwise put $c = 1/\|\mathbf{x}\|$, $d = 1/\|\mathbf{y}\|$; then $0 \le \|c\mathbf{x} \pm d\mathbf{y}\|^2 = 2 \pm 2cd\langle \mathbf{x}, \mathbf{y} \rangle$, so $|\langle \mathbf{x}, \mathbf{y} \rangle| \le 1/(cd)$.

Hence $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 \le (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$. Applying (3) to $\mathbf{x} = (\mathbf{x}-\mathbf{y}) + \mathbf{y}$ gives (3′). ■

On $\mathbb{R}^n$ we have the **euclidean norm** $\|\mathbf{x}\|$ coming from the dot product, and the **sup norm**

$$|\mathbf{x}| = \max\{|x_1|, \dots, |x_n|\}.$$

The sup norm is often more convenient. The two are related by

$$|\mathbf{x}| \le \|\mathbf{x}\| \le \sqrt{n}\,|\mathbf{x}|, \tag{1.1}$$

since $\max_i x_i^2 \le \sum_i x_i^2 \le n \max_i x_i^2$.

## 1.3 Matrices

An $n$ by $m$ matrix $A$ has entries $a_{ij}$ ($i$ = row index, $j$ = column index). The $n$ by $m$ matrices form a vector space ($\cong \mathbb{R}^{nm}$). If $A$ is $n$ by $m$ and $B$ is $m$ by $p$, the **product** $C = A \cdot B$ is the $n$ by $p$ matrix with

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

It satisfies $A\cdot(B\cdot C) = (A\cdot B)\cdot C$, $A\cdot(B+C) = A\cdot B + A\cdot C$, $(A+B)\cdot C = A\cdot C + B\cdot C$, $(cA)\cdot B = c(A\cdot B) = A\cdot(cB)$, and $I_n \cdot A = A = A \cdot I_m$ for $A$ of size $n$ by $m$, where the **identity matrix** $I_k$ has entries $\delta_{ij}$ ($\delta_{ij} = 1$ if $i = j$, 0 otherwise).

The sup norm extends to matrices: $|A| = \max\{|a_{ij}| : i = 1,\dots,n,\ j = 1,\dots,m\}$; it is a norm.

**Theorem 1.3**

If $A$ has size $n$ by $m$ and $B$ has size $m$ by $p$, then $|A\cdot B| \le m|A|\,|B|$.

**Proof [Supplement].** $|c_{ij}| = |\sum_{k=1}^m a_{ik}b_{kj}| \le \sum_{k=1}^m |A||B| = m|A||B|$ for every $i, j$. ■

## 1.4 Linear transformations

**Definition 1.5 (linear transformation)**

A function $T : V \to W$ between vector spaces is a **linear transformation** if $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ and $T(c\mathbf{x}) = cT(\mathbf{x})$. If moreover $T$ is a bijection, $T$ is a **linear isomorphism**.

Composites of linear transformations are linear, and the inverse of a linear isomorphism is linear.

**Theorem 1.4**

Let $V$ be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_m$. Let $W$ be a vector space. Given any $m$ vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ in $W$, there is exactly one linear transformation $T : V \to W$ such that $T(\mathbf{a}_i) = \mathbf{b}_i$ for all $i$.

**Proof [Supplement].** Each $\mathbf{x} \in V$ has a unique expression $\mathbf{x} = \sum c_i \mathbf{a}_i$; define $T(\mathbf{x}) = \sum c_i \mathbf{b}_i$. Uniqueness of the coefficients makes $T$ well defined, and since the coefficients of $\mathbf{x}+\mathbf{y}$ and $c\mathbf{x}$ are the sums and multiples of those of $\mathbf{x}, \mathbf{y}$, $T$ is linear. Any linear $T'$ with $T'(\mathbf{a}_i) = \mathbf{b}_i$ satisfies $T'(\sum c_i \mathbf{a}_i) = \sum c_i \mathbf{b}_i$, so $T' = T$. ■

The maps $(x_1, \dots, x_n) \mapsto [x_1 \ \cdots \ x_n]$ (**row matrix**) and $(x_1, \dots, x_n) \mapsto$ the $n$ by 1 **column matrix** with entries $x_i$ are linear isomorphisms. Representing elements of $\mathbb{R}^m$ as column matrices, a fixed

$n$ by $m$ matrix $A$ defines a linear map $T : \mathbb{R}^m \to \mathbb{R}^n$, $T(\mathbf{x}) = A \cdot \mathbf{x}$. Conversely every linear $T : \mathbb{R}^m \to \mathbb{R}^n$ has this form: let $\mathbf{b}_j = T(\mathbf{e}_j)$ and $A = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_m]$; then $A \cdot \mathbf{e}_j = \mathbf{b}_j = T(\mathbf{e}_j)$ for all $j$, so $A \cdot \mathbf{x} = T(\mathbf{x})$ by Theorem 1.4.

**Convention**

Elements of $\mathbb{R}^n$ are column matrices unless stated otherwise.

## 1.5 Rank of a matrix

For an $n$ by $m$ matrix $A$, the **column space** is the subspace of $\mathbb{R}^n$ spanned by the columns; its dimension is the **column rank** ($\leq \min(m, n)$). The **row space** is the subspace of $\mathbb{R}^m$ spanned by the rows; its dimension is the **row rank**.

**Theorem 1.5**

For any matrix $A$, the row rank of $A$ equals the column rank of $A$.

**Proof [Supplement].** Let $r$ be the column rank and let $\mathbf{c}_1, \ldots, \mathbf{c}_r \in \mathbb{R}^n$ be a basis of the column space; let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_r]$ ($n$ by $r$). Each column of $A$ is a combination of the $\mathbf{c}_k$, so $A = C \cdot D$ for some $r$ by $m$ matrix $D$. Then each row of $A$ is a combination of the $r$ rows of $D$ (row $i$ of $A$ is $\sum_k c_{ik}\cdot$(row $k$ of $D$)), so the row rank of $A$ is $\leq r$. Applying this to $A^{\mathrm{tr}}$ (whose columns are the rows of $A$) gives the reverse inequality. ■

The common value is the **rank** of $A$.

**Elementary row operations** on a matrix are:

(1) exchange rows $i_1 \neq i_2$;

(2) replace row $i_1$ by itself plus $c$ times row $i_2$ ($i_1 \neq i_2$);

(3) multiply row $i$ by a scalar $\lambda \neq 0$. Each is invertible, and its inverse is an elementary operation of the same type.

**Theorem 1.6**

If $B$ is the matrix obtained by applying an elementary row operation to $A$, then $\operatorname{rank} B = \operatorname{rank} A$.

**Proof [Supplement].** Each row of $B$ is a linear combination of rows of $A$, so the row space of $B$ is contained in that of $A$. Since $A$ is obtained from $B$ by the inverse elementary operation, the reverse inclusion holds too. The row spaces coincide, so the row ranks agree. ■

**Gauss–Jordan reduction** applies elementary operations to bring $A$ to **echelon form** (stairstep form): below the "stairsteps" all entries are 0 and the corner entries (pivots) are nonzero. Operations of types (1) and (2) suffice. The nonzero rows of a matrix in echelon form are independent, so its rank is the number of nonzero rows. Using further operations of types (2) and (3) one reaches the **reduced echelon form**, in which each corner entry is 1 and all entries directly above it are 0.

## 1.6 Transpose

The **transpose** of the $n$ by $m$ matrix $A$ is the $m$ by $n$ matrix $A^{\mathrm{tr}}$ with entries $d_{ij} = a_{ji}$. Then $(A^{\mathrm{tr}})^{\mathrm{tr}} = A$, $(A + B)^{\mathrm{tr}} = A^{\mathrm{tr}} + B^{\mathrm{tr}}$, $(A \cdot C)^{\mathrm{tr}} = C^{\mathrm{tr}} \cdot A^{\mathrm{tr}}$, and $\operatorname{rank} A^{\mathrm{tr}} = \operatorname{rank} A$ (the row rank of $A^{\mathrm{tr}}$ is the column rank of $A$).

# §2. Matrix Inversion and Determinants

**Motivation.** The determinant, characterized by its row properties, yields the product formula, the inverse formula and cofactor expansion.

## 2.1 Elementary matrices

**Definition 2.1 (elementary matrix)**

An **elementary matrix** of size $n$ by $n$ is the matrix obtained by applying one of the elementary row operations to $I_n$.

There are three types: $E$ (rows $i_1, i_2$ of $I_n$ exchanged), $E'$ ($I_n$ with an extra entry $c$ in row $i_1$, column $i_2$), and $E''$ ($I_n$ with the $(i,i)$ entry replaced by $\lambda \neq 0$).

**Theorem 2.1**

Let $A$ be an $n$ by $m$ matrix. Any elementary row operation on $A$ may be carried out by premultiplying $A$ by the corresponding elementary matrix.

**Proof.** Direct computation: multiplying $A$ on the left by $E$ interchanges rows $i_1$ and $i_2$ of $A$; by $E'$ replaces row $i_1$ by itself plus $c$ times row $i_2$; by $E''$ multiplies row $i$ by $\lambda$. ■

## 2.2 The inverse of a matrix

**Definition 2.2 (left and right inverse)**

Let $A$ be $n$ by $m$, and $B, C$ be $m$ by $n$. $B$ is a **left inverse** for $A$ if $B \cdot A = I_m$; $C$ is a **right inverse** for $A$ if $A \cdot C = I_n$.

**Theorem 2.2**

If $A$ has both a left inverse $B$ and a right inverse $C$, then they are unique and equal.

**Proof.** $C = I_m \cdot C = (B \cdot A) \cdot C = B \cdot (A \cdot C) = B \cdot I_n = B$. If $B_1$ is another left inverse, the same computation gives $C = B_1$, so $B_1 = B$; similarly $C$ is unique. ■

**Definition 2.3 (invertible)**

If $A$ has both a left and a right inverse, $A$ is **invertible**; the unique two-sided inverse is denoted $A^{-1}$.

**Theorem 2.3**

Let $A$ be a matrix of size $n$ by $m$. If $A$ is invertible, then $n = m = \operatorname{rank} A$.

**Proof.**

*Step 1.* For any $k$ by $n$ matrix $D$, $\operatorname{rank}(D \cdot A) \leq \operatorname{rank} A$. Indeed, if $R$ is a 1 by $n$ row matrix, $R \cdot A$ is a linear combination of the rows of $A$. Each row of $D \cdot A$ is (row of $D$)$\cdot A$, so the row space of $D \cdot A$ lies in the row space of $A$.

*Step 2.* If $A$ has a left inverse $B$, then $\operatorname{rank} A = m$ (the number of columns). Indeed $m = \operatorname{rank} I_m = \operatorname{rank}(B \cdot A) \le \operatorname{rank} A$ by Step 1, while the row space of $A$ is a subspace of $\mathbb{R}^m$, so $\operatorname{rank} A \le m$.

*Step 3.* Let $B = A^{-1}$. Since $B$ is a left inverse, $\operatorname{rank} A = m$ by Step 2. Since $B$ is a right inverse, $B^{\mathrm{tr}} \cdot A^{\mathrm{tr}} = I_n^{\mathrm{tr}} = I_n$, so $B^{\mathrm{tr}}$ is a left inverse of the $m$ by $n$ matrix $A^{\mathrm{tr}}$ and Step 2 gives $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{tr}} = n$. ■

**Theorem 2.4**

Let $A$ be a matrix of size $n$ by $m$ with $n = m = \operatorname{rank} A$. Then $A$ is invertible; furthermore, $A$ equals a product of elementary matrices.

**Proof.**

*Step 1.* Every elementary matrix is invertible with elementary inverse: $E$ is its own inverse, $E'$ with $c$ replaced by $-c$ inverts $E'$, and $E''$ with $\lambda$ replaced by $1/\lambda$ inverts $E''$.

*Step 2.* Reduce $A$ to reduced echelon form $C$ by elementary operations. $C$ is square of rank $n$, so it has $n$ nonzero rows; hence $C = I_n$. By Theorem 2.1 there are elementary matrices $E_1, \dots, E_k$ with $E_k \cdots E_2 E_1 A = I_n$. Multiplying on the left successively by $E_k^{-1}, \dots, E_1^{-1}$ gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

a product of elementary matrices, and $B = E_k E_{k-1} \cdots E_1$ is both a left and a right inverse for $A$. ■

**Theorem 2.5**

If $A$ is a square matrix and $B$ is a left inverse for $A$, then $B$ is also a right inverse for $A$.

**Proof.** By Step 2 of the proof of Theorem 2.3, $\operatorname{rank} A$ equals the number of columns, which (as $A$ is square) equals the number of rows. By Theorem 2.4, $A$ has an inverse, which equals $B$ by Theorem 2.2. ■

An $n$ by $n$ matrix is **singular** if $\operatorname{rank} A < n$, **non-singular** otherwise. By the above, $A$ is invertible iff it is non-singular.

## 2.3 Determinants

We write $\det A$ ($|A|$ is the sup norm) and characterize it by axioms; existence for all $n$ is proved in §27.

**Definition 2.4 (determinant function)**

A function assigning to each $n$ by $n$ matrix $A$ a real number $\det A$ is a **determinant function** if:

(1) if $B$ is obtained from $A$ by exchanging two rows, then $\det B = -\det A$;

(2) for each $i$, $\det A$ is linear as a function of the $i$th row alone: if $A_i(\mathbf{x})$ denotes $A$ with row $i$ replaced by $\mathbf{x}$, then $\det A_i(a\mathbf{x} + b\mathbf{y}) = a \det A_i(\mathbf{x}) + b \det A_i(\mathbf{y})$;

(3) $\det I_n = 1$.

**Example 2.1**

For 1 by 1 matrices $\det[a] = a$; for 2 by 2 matrices $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$; for 3 by 3,

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1 - a_1b_3c_2 - a_2b_1c_3.$$

These satisfy the axioms. For larger matrices a direct formula has $n!$ terms (24 for $n = 4$, 120 for $n = 5$), so a less direct approach is needed.

**Theorem 2.6**

Let $A$ be an $n$ by $n$ matrix.

(a) If $E$ is the elementary matrix exchanging rows $i_1$ and $i_2$, then $\det(E \cdot A) = -\det A$.

(b) If $E'$ replaces row $i_1$ of $A$ by itself plus $c$ times row $i_2$, then $\det(E' \cdot A) = \det A$.

(c) If $E''$ multiplies row $i$ of $A$ by $\lambda \neq 0$, then $\det(E'' \cdot A) = \lambda \det A$.

(d) If $A = I_n$, then $\det A = 1$.

**Proof.** (a) is Axiom 1 and (d) is Axiom 3;

(c) is the special case $\det A_i(\lambda \mathbf{x}) = \lambda \det A_i(\mathbf{x})$ of Axiom 2. For (b): if $A$ has two equal rows, exchanging them does not change $A$ but changes the sign of $\det A$, so $\det A = 0$. Let $i = i_1$, $\mathbf{x} =$ row $i_1$, $\mathbf{y} =$ row $i_2$. Then

$$\det(E' \cdot A) = \det A_i(\mathbf{x} + c\mathbf{y}) = \det A_i(\mathbf{x}) + c \det A_i(\mathbf{y}) = \det A_i(\mathbf{x}) = \det A,$$

since $A_i(\mathbf{y})$ has two equal rows. ■

Taking $A = I_n$: $\det E = -1$, $\det E' = 1$, $\det E'' = \lambda$.

**Theorem 2.7**

Let $A$ be a square matrix. If the rows of $A$ are independent, then $\det A \neq 0$; if the rows are dependent, then $\det A = 0$. Thus an $n$ by $n$ matrix $A$ has rank $n$ if and only if $\det A \neq 0$.

**Proof.** If row $i$ of $A$ is zero, then $\det A = 0$: multiplying that row by 2 leaves $A$ unchanged but doubles $\det A$. An elementary operation multiplies the determinant by $-1$, 1 or $\lambda \neq 0$, so it does not affect whether $\det A$ vanishes. Reduce $A$ to echelon form $B$ (types (1), (2)). If the rows of $A$ are dependent, $\operatorname{rank} B = \operatorname{rank} A < n$, so $B$ has a zero row, $\det B = 0$, and $\det A = 0$. If they are independent, reduce further to reduced echelon form $C$; being square of rank $n$, $C = I_n$, so $\det C = 1 \neq 0$ and $\det A \neq 0$. ■

**Theorem 2.8**

Given a square matrix $A$, reduce it to echelon form $B$ by elementary row operations of types (1) and (2). If $B$ has a zero row, then $\det A = 0$. Otherwise, let $k$ be the number of row exchanges involved; then $\det A = (-1)^k$ times the product of the diagonal entries of $B$.

**Proof.** If $B$ has a zero row, $\operatorname{rank} A < n$ and $\det A = 0$. Otherwise $\det A = (-1)^k \det B$ by Theorem 2.6(a),(b), and $B$ is upper triangular with nonzero diagonal entries $b_{11}, \ldots, b_{nn}$. Using type (2) operations, clear the entries above the diagonal; the diagonal is unaffected, giving a diagonal matrix

$C$ with $\det C = \det B$. Multiplying row $i$ of $C$ by $1/b_{ii}$ for each $i$ yields $I_n$, so by Theorem 2.6(c),(d), $1 = \det I_n = (1/b_{11}) \cdots (1/b_{nn}) \det C$. Hence $\det B = \det C = b_{11} \cdots b_{nn}$. ■

**Corollary 2.9**

The determinant function is uniquely characterized by its three axioms. It is also characterized by the four properties listed in Theorem 2.6.

**Proof.** The calculation of $\det A$ in Theorem 2.8 uses only properties (a)–(d) of Theorem 2.6, which follow from the axioms. ■

**Theorem 2.10**

Let $A$ and $B$ be $n$ by $n$ matrices. Then $\det(A \cdot B) = (\det A) \cdot (\det B)$.

**Proof.**

*Step 1.* The theorem holds when $A$ is elementary: $\det(E \cdot B) = -\det B = (\det E)(\det B)$, $\det(E' \cdot B) = \det B = (\det E')(\det B)$, $\det(E'' \cdot B) = \lambda \det B = (\det E'')(\det B)$.

*Step 2.* If $\operatorname{rank} A = n$, $A = E_1 \cdots E_k$ is a product of elementary matrices (Theorem 2.4), and repeated use of Step 1 gives $\det(A \cdot B) = (\det E_1) \cdots (\det E_k)(\det B)$ for all $B$. With $B = I_n$ this reads $\det A = (\det E_1) \cdots (\det E_k)$, and the theorem follows.

*Step 3.* If $\operatorname{rank} A < n$, then $\operatorname{rank}(A \cdot B) = \operatorname{rank}(B^{\mathrm{tr}} \cdot A^{\mathrm{tr}}) \leq \operatorname{rank} A^{\mathrm{tr}} < n$ by Step 1 of Theorem 2.3, so both sides vanish by Theorem 2.7. ■

**Theorem 2.11**

$\det A^{\mathrm{tr}} = \det A$.

**Proof.**

*Step 1.* For elementary matrices: $E^{\mathrm{tr}} = E$, $(E'')^{\mathrm{tr}} = E''$, and $(E')^{\mathrm{tr}}$ is again of type (2), so both have determinant 1.

*Step 2.* If $\operatorname{rank} A = n$, write $A = E_1 \cdots E_k$. Then by Theorem 2.10 and Step 1,

$$\det A^{\mathrm{tr}} = \det(E_k^{\mathrm{tr}} \cdots E_1^{\mathrm{tr}}) = (\det E_k^{\mathrm{tr}}) \cdots (\det E_1^{\mathrm{tr}}) = (\det E_1) \cdots (\det E_k) = \det A.$$

*Step 3.* If $\operatorname{rank} A < n$, then $\operatorname{rank} A^{\mathrm{tr}} < n$ and both determinants are 0. ■

## 2.4 A formula for $A^{-1}$

**Definition 2.5 (minor, cofactor)**

For an $n$ by $n$ matrix $A$, the $(n-1)$ by $(n-1)$ matrix $A_{ij}$ obtained by deleting row $i$ and column $j$ is the $(i,j)$**-minor**; the number $(-1)^{i+j} \det A_{ij}$ is the $(i,j)$**-cofactor** of $A$.

**Lemma 2.12**

Let $A$ be an $n$ by $n$ matrix and let $b$ denote its entry in row $i$ and column $j$.

(a) If all the entries in row $i$ other than $b$ vanish, then $\det A = b(-1)^{i+j} \det A_{ij}$.

(b) The same holds if all the entries in column $j$ other than $b$ vanish.

**Proof.**

*Step 1.* Fix $b, a_2, \dots, a_n$. For an $(n-1)$ by $(n-1)$ matrix $D$ let $A(D)$ be the $n$ by $n$ matrix with first row $(b, a_2, \dots, a_n)$, first column $(b, 0, \dots, 0)^{\mathrm{tr}}$ and lower right block $D$. We show $\det A(D) = b \det D$. If $b = 0$, $\operatorname{rank} A(D) < n$ and both sides vanish. If $b \neq 0$, put $f(D) = (1/b) \det A(D)$. Exchanging two rows of $D$, adding a multiple of one row of $D$ to another, or multiplying a row of $D$ by $\lambda$ has the same effect on the corresponding rows $(i+1)$ of $A(D)$, so $f$ changes by $-1$, $1$, $\lambda$, respectively. If $D = I_{n-1}$, $A(D)$ is in echelon form and $\det A(D) = b$ by Theorem 2.8, so $f(I_{n-1}) = 1$. By Corollary 2.9, $f(D) = \det D$.

*Step 2.* Taking transposes (Theorem 2.11), the same formula holds for the matrix with first row $(b, 0, \dots, 0)$, first column $(b, a_2, \dots, a_n)^{\mathrm{tr}}$ and block $D$.

*Step 3.* In the general case, $i-1$ exchanges of adjacent rows bring row $i$ to the top without changing the order of the other rows; then $j-1$ exchanges of adjacent columns bring column $j$ to the left edge. The resulting matrix $C$ is of the form in Step 1 or 2, and its $(1,1)$-minor $C_{11}$ equals $A_{ij}$. Each row exchange, and (by Theorem 2.11) each column exchange, changes the sign of the determinant, so

$$\det A = (-1)^{(i-1)+(j-1)} \det C = (-1)^{i+j} b \det C_{11} = (-1)^{i+j} b \det A_{ij}.$$

■

**Theorem 2.13 (Cramer's rule)**

Let $A$ be an $n$ by $n$ matrix with successive columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{c} = (c_1, \dots, c_n)$ be column matrices. If $A \cdot \mathbf{x} = \mathbf{c}$, then

$$(\det A) \cdot x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n].$$

**Proof.** Let $C = [\mathbf{e}_1 \cdots \mathbf{e}_{i-1}\ \mathbf{x}\ \mathbf{e}_{i+1} \cdots \mathbf{e}_n]$. Since $A \cdot \mathbf{e}_j = \mathbf{a}_j$ and $A \cdot \mathbf{x} = \mathbf{c}$, we have $A \cdot C = [\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$, so by Theorem 2.10, $(\det A)(\det C)$ equals the right-hand side. $C$ is $I_n$ except for column $i$, which is $\mathbf{x}$; expanding by Lemma 2.12 along row $i$ (whose only nonzero entry may be $x_i$ in column $i$) gives $\det C = x_i (-1)^{i+i} \det I_{n-1} = x_i$. ■

**Theorem 2.14**

Let $A$ be an $n$ by $n$ matrix of rank $n$; let $B = A^{-1}$. Then

$$b_{ij} = \frac{(-1)^{j+i} \det A_{ji}}{\det A}.$$

**Proof.** Fix $j$ and let $\mathbf{x}$ be the $j$th column of $B$. Then $A \cdot B = I_n$ gives $A \cdot \mathbf{x} = \mathbf{e}_j$, and Cramer's rule gives $(\det A) x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{e}_j\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$. Column $i$ of this matrix has a single nonzero entry $1$ in row $j$, so by Lemma 2.12(b) the determinant is $1 \cdot (-1)^{j+i} \det A_{ji}$. Since $x_i = b_{ij}$, the theorem follows. ■

This yields the algorithm:

(1) form the **matrix of cofactors**, with $(i,j)$ entry $(-1)^{i+j} \det A_{ij}$;

(2) transpose it;

(3) divide each entry by $\det A$. It shows that the entries of $A^{-1}$ are continuous functions of those of $A$; Gauss–Jordan reduction is better for computation.

### 2.5 Expansion by cofactors

**Theorem 2.15**

Let $A$ be an $n$ by $n$ matrix and fix $i$. Then

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det A_{ik}.$$

**Proof.** Let $A_i(\mathbf{x})$ be $A$ with row $i$ replaced by the row matrix $\mathbf{x}$. Row $i$ of $A$ is $\sum_k a_{ik}\mathbf{e}_k$, so by linearity (Axiom 2) and Lemma 2.12,

$$\det A = \sum_{k=1}^{n} a_{ik} \det A_i(\mathbf{e}_k) = \sum_{k=1}^{n} a_{ik}(-1)^{i+k} \det A_{ik}.$$

■

This is the expansion of the determinant by cofactors of the $i$th row; taking transposes gives expansion by cofactors of the $j$th column.

## §3. Review of Topology in $\mathbb{R}^n$

**Motivation.** Limits need open sets, closed sets and continuity; the metric-space language is reused for manifolds.

### 3.1 Metric spaces

**Definition 3.1 (metric space)**

A **metric** on a set $X$ is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:
(1) $d(x, y) = d(y, x)$;
(2) $d(x, y) \geq 0$, with equality iff $x = y$;
(3) $d(x, z) \leq d(x, y) + d(y, z)$. A **metric space** is a set with a specific metric. A subset $Y \subset X$ with the restricted metric is a **subspace** of $X$.

On $\mathbb{R}^n$ we have the **euclidean metric** $\|\mathbf{x} - \mathbf{y}\|$ and the **sup metric** $|\mathbf{x} - \mathbf{y}|$. $\mathbb{R}^n$ is called $n$-dimensional **euclidean space**.

**Definition 3.2 (open and closed sets)**

For $x_0 \in X$ and $\epsilon > 0$, the **$\epsilon$-neighborhood** of $x_0$ is $U(x_0; \epsilon) = \{x : d(x, x_0) < \epsilon\}$. A set $U \subset X$ is **open** in $X$ if for each $x_0 \in U$ some $U(x_0; \epsilon) \subset U$; a set $C$ is **closed** in $X$ if $X - C$ is open. Any open set containing $x_0$ is a **neighborhood** of $x_0$.

By the triangle inequality, an $\epsilon$-neighborhood is open: if $d(x, x_0) < \epsilon$, then $U(x; \epsilon - d(x, x_0)) \subset U(x_0; \epsilon)$.

**Theorem 3.1**

Let $(X, d)$ be a metric space. Then finite intersections and arbitrary unions of open sets of $X$ are open in $X$. Similarly, finite unions and arbitrary intersections of closed sets of $X$ are closed in $X$.

**Proof [Supplement].** If $x$ lies in a union of open sets, it lies in one of them, $U_\alpha$, and some $U(x;\epsilon) \subset U_\alpha$ is contained in the union. If $x \in U_1 \cap \cdots \cap U_k$, choose $\epsilon_i$ with $U(x;\epsilon_i) \subset U_i$ and let $\epsilon = \min \epsilon_i > 0$. The statements about closed sets follow by De Morgan's laws. ■

**Theorem 3.2**

Let $X$ be a metric space and $Y$ a subspace. A subset $A$ of $Y$ is open in $Y$ if and only if $A = U \cap Y$ for some $U$ open in $X$. Similarly, $A$ is closed in $Y$ if and only if $A = C \cap Y$ for some $C$ closed in $X$.

**Proof [Supplement].** The $\epsilon$-neighborhood of $y \in Y$ in $Y$ is $U_X(y;\epsilon) \cap Y$. If $U$ is open in $X$ and $y \in U \cap Y$, then $U_X(y;\epsilon) \subset U$ for some $\epsilon$, so $U_Y(y;\epsilon) \subset U \cap Y$. Conversely, if $A$ is open in $Y$, choose for each $y \in A$ an $\epsilon_y$ with $U_X(y;\epsilon_y) \cap Y \subset A$ and put $U = \bigcup_{y\in A} U_X(y;\epsilon_y)$; then $U$ is open in $X$ and $U \cap Y = A$. For closed sets take complements: $Y - A = (X - C) \cap Y$. ■

Hence if $A$ is open in $Y$ and $Y$ is open in $X$, then $A$ is open in $X$; likewise for closed sets.

**Definition 3.3 (limit point, closure)**

$x_0 \in X$ is a **limit point** of $A \subset X$ if every $\epsilon$-neighborhood of $x_0$ meets $A$ in a point other than $x_0$ (equivalently, in infinitely many points of $A$). The **closure** $\bar{A}$ of $A$ is $A$ together with all its limit points.

**Theorem 3.3**

If $A$ is a subset of $X$, then $\bar{A}$ is a closed set of $X$. A subset of $X$ is closed if and only if it contains all its limit points.

**Proof [Supplement].** A set $C$ is closed iff every $x \notin C$ has a neighborhood disjoint from $C$, i.e. iff no point outside $C$ is a limit point of $C$; this is the second statement. For the first: if $x \notin \bar{A}$, some $U(x;\epsilon)$ misses $A$. This open set also misses every limit point $y$ of $A$: a point $y \in U(x;\epsilon)$ has a neighborhood inside $U(x;\epsilon)$, which contains no point of $A$. So $U(x;\epsilon) \cap \bar{A} = \emptyset$ and $X - \bar{A}$ is open. ■

In $\mathbb{R}^n$, the $\epsilon$-neighborhood of $\mathbf{a}$ in the euclidean metric is the **open ball** $B(\mathbf{a};\epsilon)$; in the sup metric it is the **open cube** $C(\mathbf{a};\epsilon)$. By (1.1),

$$B(\mathbf{a};\epsilon) \subset C(\mathbf{a};\epsilon) \subset B(\mathbf{a};\sqrt{n}\,\epsilon).$$

**Theorem 3.4**

If $X$ is a subspace of $\mathbb{R}^n$, the collection of open sets of $X$ is the same whether one uses the euclidean metric or the sup metric on $X$. The same is true for the collection of closed sets of $X$.

**Proof [Supplement].** If $A$ is open for the sup metric and $\mathbf{x} \in A$, then $C(\mathbf{x};\epsilon) \cap X \subset A$ for some $\epsilon$, hence $B(\mathbf{x};\epsilon) \cap X \subset A$. If $A$ is open for the euclidean metric, $B(\mathbf{x};\epsilon) \cap X \subset A$, hence $C(\mathbf{x};\epsilon/\sqrt{n}) \cap X \subset A$. Closed sets are the complements. ■

A property of $X$ depending only on its collection of open sets is a **topological property**; limits, continuity and compactness are examples.

## 3.2 Limits and continuity

**Definition 3.4 (continuity)**

Let $X, Y$ be metric spaces. $f : X \to Y$ is **continuous at** $x_0$ if for each open $V \ni f(x_0)$ there is an open $U \ni x_0$ with $f(U) \subset V$; equivalently, for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), f(x_0)) < \epsilon$ whenever $d_X(x, x_0) < \delta$. $f$ is **continuous** if it is continuous at every point; equivalently, $f^{-1}(V)$ is open in $X$ for each open $V$ of $Y$ (or $f^{-1}(D)$ is closed for each closed $D$).

If some $\delta$-neighborhood of $x_0$ is $\{x_0\}$, then $x_0$ is an **isolated point** of $X$, and every $f$ is continuous at $x_0$. Constant maps and the identity are continuous.

**Theorem 3.5**

(a) Let $x_0 \in A$, where $A$ is a subspace of $X$. If $f : X \to Y$ is continuous at $x_0$, then $f|A : A \to Y$ is continuous at $x_0$.
(b) Let $f : X \to Y$ and $g : Y \to Z$. If $f$ is continuous at $x_0$ and $g$ is continuous at $y_0 = f(x_0)$, then $g \circ f : X \to Z$ is continuous at $x_0$.

**Proof [Supplement].** (a) The same $\delta$ works for points of $A$.

(b) Given $\epsilon$, choose $\eta$ with $d_Z(g(y), g(y_0)) < \epsilon$ for $d_Y(y, y_0) < \eta$, then $\delta$ with $d_Y(f(x), y_0) < \eta$ for $d_X(x, x_0) < \delta$. ■

**Theorem 3.6**

(a) Let $X$ be a metric space and $f : X \to \mathbb{R}^n$, $f(x) = (f_1(x), \ldots, f_n(x))$. Then $f$ is continuous at $x_0$ if and only if each **component function** $f_i : X \to \mathbb{R}$ is continuous at $x_0$.
(b) Let $f, g : X \to \mathbb{R}$ be continuous at $x_0$. Then $f + g$, $f - g$ and $f \cdot g$ are continuous at $x_0$, and $f/g$ is continuous at $x_0$ if $g(x_0) \neq 0$.
(c) The projection $\pi_i : \mathbb{R}^n \to \mathbb{R}$, $\pi_i(\mathbf{x}) = x_i$, is continuous.

**Proof [Supplement].** (a) Use the sup metric: $|f(x) - f(x_0)| < \epsilon$ iff $|f_i(x) - f_i(x_0)| < \epsilon$ for all $i$; take the minimum of the $n$ values of $\delta$.

(c) $|\pi_i(\mathbf{x}) - \pi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$.

(b) Addition, subtraction and multiplication $\mathbb{R}^2 \to \mathbb{R}$ are continuous, and so is division on $\{(s, t) : t \neq 0\}$, by the usual estimates, e.g. $|st - s_0t_0| \leq |s||t - t_0| + |t_0||s - s_0|$. Compose with $x \mapsto (f(x), g(x))$, continuous by (a), and use Theorem 3.5. ■

E.g. $f(s, t, u, v) = \sin(s + t)/e^{uv}$ is continuous on $\mathbb{R}^4$.

**Definition 3.5 (limit)**

Let $A \subset X$, $f : A \to Y$, and let $x_0$ be a limit point of $A$ (not necessarily in $A$). We say $f(x) \to y_0$ as $x \to x_0$, or $\lim_{x \to x_0} f(x) = y_0$, if for each open $V \ni y_0$ there is an open $U \ni x_0$ with $f(x) \in V$ for all $x \in U \cap A$, $x \neq x_0$. Equivalently: for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), y_0) < \epsilon$ whenever $x \in A$ and $0 < d_X(x, x_0) < \delta$.

The value of $f$ at $x_0$ plays no role.

**Theorem 3.7**

Let $f : X \to Y$. If $x_0$ is an isolated point of $X$, then $f$ is continuous at $x_0$. Otherwise, $f$ is continuous at $x_0$ if and only if $f(x) \to f(x_0)$ as $x \to x_0$.

**Proof [Supplement].** Compare the $\epsilon$-$\delta$ conditions: they differ only in whether $x = x_0$ is allowed, and at $x = x_0$ the condition $d_Y(f(x), f(x_0)) < \epsilon$ holds trivially. ■

**Theorem 3.8**

(a) Let $A \subset X$ and $f : A \to \mathbb{R}^n$, $f = (f_1, \ldots, f_n)$; let $\mathbf{a} = (a_1, \ldots, a_n)$. Then $f(x) \to \mathbf{a}$ as $x \to x_0$ if and only if $f_i(x) \to a_i$ for each $i$.
(b) Let $f, g : A \to \mathbb{R}$. If $f(x) \to a$ and $g(x) \to b$ as $x \to x_0$, then $f+g \to a+b$, $f-g \to a-b$, $f \cdot g \to a \cdot b$, and $f/g \to a/b$ if $b \neq 0$.

**Proof [Supplement].** Extend $f, g$ to $A \cup \{x_0\}$ by $f(x_0) = \mathbf{a}$ (resp. $a$, $b$). The limit statements become continuity at $x_0$ of the extended functions (Theorem 3.7), and Theorem 3.6 applies. ■

### 3.3 Interior and exterior

**Definition 3.6 (interior, exterior, boundary)**

For $A \subset \mathbb{R}^n$: $\operatorname{Int} A$ is the union of all open sets of $\mathbb{R}^n$ contained in $A$; $\operatorname{Ext} A$ is the union of all open sets disjoint from $A$; $\operatorname{Bd} A$ consists of the points in neither $\operatorname{Int} A$ nor $\operatorname{Ext} A$.

A point $\mathbf{x}$ is in $\operatorname{Bd} A$ iff every open set containing $\mathbf{x}$ meets both $A$ and $\mathbb{R}^n - A$. $\mathbb{R}^n$ is the disjoint union of $\operatorname{Int} A$, $\operatorname{Ext} A$ and $\operatorname{Bd} A$; the first two are open and $\operatorname{Bd} A$ is closed.

For a **rectangle** $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ one checks $\operatorname{Int} Q = (a_1, b_1) \times \cdots \times (a_n, b_n)$ (an **open rectangle**), $\operatorname{Ext} Q = \mathbb{R}^n - Q$, $\operatorname{Bd} Q = Q - \operatorname{Int} Q$. The open cube $C(\mathbf{a}; \epsilon) = (a_1 - \epsilon, a_1 + \epsilon) \times \cdots \times (a_n - \epsilon, a_n + \epsilon)$ is an open rectangle; the corresponding closed rectangle $[a_1 - \epsilon, a_1 + \epsilon] \times \cdots \times [a_n - \epsilon, a_n + \epsilon]$ is the **closed cube** (or simply cube) centered at $\mathbf{a}$.

## §4. Compact Subspaces and Connected Subspaces of $\mathbb{R}^n$

**Motivation.** Compactness turns pointwise statements into uniform ones (uniform continuity, maxima, Lebesgue numbers).

### 4.1 Compact spaces

**Definition 4.1 (compact)**

Let $X \subset \mathbb{R}^n$. A **covering** of $X$ is a collection of subsets of $\mathbb{R}^n$ whose union contains $X$; it is an **open covering** if each member is open in $\mathbb{R}^n$. $X$ is **compact** if every open covering of $X$ contains a finite subcollection that also covers $X$.

**Theorem 4.1**

A subspace $X$ of $\mathbb{R}^n$ is compact if and only if for every collection of sets open in $X$ whose union is $X$, there is a finite subcollection whose union equals $X$.

**Proof.** Suppose $X$ is compact and $\{A_\alpha\}$ are open in $X$ with union $X$. By Theorem 3.2 choose $U_\alpha$ open in $\mathbb{R}^n$ with $A_\alpha = U_\alpha \cap X$. The $U_\alpha$ cover $X$, so finitely many, $U_{\alpha_1}, \ldots, U_{\alpha_k}$, do; then $A_{\alpha_1}, \ldots, A_{\alpha_k}$ have union $X$. Conversely, given an open covering $\{U_\alpha\}$ of $X$ by open sets of $\mathbb{R}^n$, the sets $U_\alpha \cap X$ are open in $X$ with union $X$; a finite subfamily with union $X$ gives finitely many $U_\alpha$ covering $X$. ■

**Theorem 4.2**

The subspace $[a, b]$ of $\mathbb{R}$ is compact.

**Proof [Supplement].** Let $\mathcal{A}$ be an open covering of $[a, b]$ and let $S$ be the set of $x \in [a, b]$ such that $[a, x]$ is covered by finitely many members of $\mathcal{A}$. Then $a \in S$; let $c = \sup S$. Some $U \in \mathcal{A}$ contains $c$, and with it an interval $(c - \delta, c + \delta)$. There is $x \in S$ with $x > c - \delta$; adding $U$ to the finite family covering $[a, x]$ covers $[a, \min(c + \delta/2, b)]$. Hence $c \in S$, and if $c < b$ we would get a point of $S$ larger than $c$. So $c = b \in S$. ■

**Definition 4.2 (bounded)**

$X \subset \mathbb{R}^n$ is **bounded** if there is $M$ with $|\mathbf{x}| \leq M$ for all $\mathbf{x} \in X$.

**Theorem 4.3**

If $X$ is a compact subspace of $\mathbb{R}^n$, then $X$ is closed and bounded.

**Proof.**

*Step 1.* Bounded: the open cubes $U_N = C(\mathbf{0}; N)$, $N = 1, 2, \ldots$, are open, increasing, and cover $\mathbb{R}^n$, hence $X$. Finitely many, $U_{N_1}, \ldots, U_{N_k}$, cover $X$; with $M = \max N_i$, $X \subset U_M$.

*Step 2.* Closed: let $\mathbf{a} \notin X$; we find a neighborhood of $\mathbf{a}$ disjoint from $X$. Let $C_N = \{\mathbf{x} : |\mathbf{x} - \mathbf{a}| \leq 1/N\}$ and $V_N = \mathbb{R}^n - C_N$. The $V_N$ are open, increasing, and cover $\mathbb{R}^n - \{\mathbf{a}\} \supset X$. Finitely many cover $X$, so $X \subset V_M$ for some $M$; then $C_M$ is disjoint from $X$, and in particular the open cube $C(\mathbf{a}; 1/M)$ lies in $\mathbb{R}^n - X$. ■

**Corollary 4.4**

Let $X$ be a compact subspace of $\mathbb{R}$. Then $X$ has a largest element and a smallest element.

**Proof.** $X$ is bounded, so it has a supremum and an infimum; $X$ is closed, so these (being limit points of $X$ or points of $X$) belong to $X$. ■

**Theorem 4.5 (Extreme-value theorem)**

Let $X$ be a compact subspace of $\mathbb{R}^m$. If $f : X \to \mathbb{R}^n$ is continuous, then $f(X)$ is a compact subspace of $\mathbb{R}^n$. In particular, if $\phi : X \to \mathbb{R}$ is continuous, then $\phi$ has a maximum value and a minimum value.

**Proof.** Let $\{V_\alpha\}$ be an open covering of $f(X)$. The sets $f^{-1}(V_\alpha)$ are open in $X$ and cover $X$, so by Theorem 4.1 finitely many, for $\alpha = \alpha_1, \ldots, \alpha_k$, cover $X$; then $V_{\alpha_1}, \ldots, V_{\alpha_k}$ cover $f(X)$. If $\phi : X \to \mathbb{R}$ is continuous, $\phi(X)$ is compact, and by Corollary 4.4 it has a largest and a smallest element. ■

**Definition 4.3 ($\epsilon$-neighborhood of a set)**

For $X \subset \mathbb{R}^n$ and $\epsilon > 0$, the union of the balls $B(\mathbf{a};\epsilon)$, $\mathbf{a} \in X$, is the **$\epsilon$-neighborhood** of $X$ in the euclidean metric; the union of the cubes $C(\mathbf{a};\epsilon)$ is the $\epsilon$-neighborhood in the sup metric.

**Theorem 4.6 (The $\epsilon$-neighborhood theorem)**

Let $X$ be a compact subspace of $\mathbb{R}^n$; let $U$ be an open set of $\mathbb{R}^n$ containing $X$. Then there is an $\epsilon > 0$ such that the $\epsilon$-neighborhood of $X$ (in either metric) is contained in $U$.

**Proof.** The euclidean $\epsilon$-neighborhood lies inside the sup-metric one, so it suffices to treat the sup metric.

*Step 1.* For $C \subset \mathbb{R}^n$ define $d(\mathbf{x}, C) = \inf\{|\mathbf{x} - \mathbf{c}| : \mathbf{c} \in C\}$, the **distance from x to** $C$. It is continuous: for $\mathbf{c} \in C$, $d(\mathbf{x}, C) - |\mathbf{x} - \mathbf{y}| \le |\mathbf{x} - \mathbf{c}| - |\mathbf{x} - \mathbf{y}| \le |\mathbf{y} - \mathbf{c}|$; taking the infimum over $\mathbf{c}$, $d(\mathbf{x}, C) - d(\mathbf{y}, C) \le |\mathbf{x} - \mathbf{y}|$, and symmetrically.

*Step 2.* Let $f(\mathbf{x}) = d(\mathbf{x}, \mathbb{R}^n - U)$ for $\mathbf{x} \in X$. Then $f$ is continuous and positive on $X$: if $\mathbf{x} \in X$, some $C(\mathbf{x};\delta) \subset U$, so $f(\mathbf{x}) \ge \delta$. Since $X$ is compact, $f$ has a minimum value $\epsilon > 0$ (Theorem 4.5). Then for $\mathbf{a} \in X$, every point $\mathbf{y}$ with $|\mathbf{y} - \mathbf{a}| < \epsilon$ lies in $U$; i.e. the $\epsilon$-neighborhood of $X$ lies in $U$. ■

Some hypothesis on $X$ is needed: if $X$ is the $x$-axis in $\mathbb{R}^2$ and $U = \{(x, y) : y^2 < 1/(1 + x^2)\}$, no $\epsilon$-neighborhood of $X$ lies in $U$.

**Theorem 4.7 (Uniform continuity)**

Let $X$ be a compact subspace of $\mathbb{R}^m$; let $f : X \to \mathbb{R}^n$ be continuous. Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $\mathbf{x}, \mathbf{y} \in X$,

$$|\mathbf{x} - \mathbf{y}| < \delta \quad \text{implies} \quad |f(\mathbf{x}) - f(\mathbf{y})| < \epsilon.$$

This result also holds if one uses the euclidean metric instead of the sup metric.

The conclusion is the condition of **uniform continuity**.

**Proof.** Consider $X \times X \subset \mathbb{R}^m \times \mathbb{R}^m$ and the **diagonal** $\Delta = \{(\mathbf{x}, \mathbf{x}) : \mathbf{x} \in X\}$, which is compact as the image of $X$ under the continuous map $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{x})$. First use the euclidean metric. The function $g : X \times X \to \mathbb{R}$, $g(\mathbf{x}, \mathbf{y}) = \|f(\mathbf{x}) - f(\mathbf{y})\|$, is continuous, so $\{(\mathbf{x}, \mathbf{y}) : g(\mathbf{x}, \mathbf{y}) < \epsilon\}$ is open in $X \times X$ and contains $\Delta$; hence it equals $U \cap (X \times X)$ for some open $U \subset \mathbb{R}^{2m}$ containing $\Delta$ (Theorem 3.2). By Theorem 4.6, some $\delta$-neighborhood of $\Delta$ lies in $U$. If $\mathbf{x}, \mathbf{y} \in X$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$, then $\|(\mathbf{x}, \mathbf{y}) - (\mathbf{y}, \mathbf{y})\| = \|(\mathbf{x} - \mathbf{y}, \mathbf{0})\| = \|\mathbf{x} - \mathbf{y}\| < \delta$, so $(\mathbf{x}, \mathbf{y})$ lies in the $\delta$-neighborhood of $\Delta$, hence in $U$, and $g(\mathbf{x}, \mathbf{y}) < \epsilon$.

For the sup metric: if $|\mathbf{x} - \mathbf{y}| < \delta/\sqrt{m}$ then $\|\mathbf{x} - \mathbf{y}\| < \delta$, whence $|f(\mathbf{x}) - f(\mathbf{y})| \le \|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon$. ■

**Lemma 4.8**

The rectangle $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ in $\mathbb{R}^n$ is compact.

**Proof.** Induction on $n$; the case $n = 1$ is Theorem 4.2. Write $Q = X \times [a_n, b_n]$ with $X$ a rectangle in $\mathbb{R}^{n-1}$, compact by the induction hypothesis. Let $\mathcal{A}$ be an open covering of $Q$.

*Step 1.* Given $t \in [a_n, b_n]$, there is $\epsilon > 0$ such that $X \times (t - \epsilon, t + \epsilon)$ is covered by finitely many elements of $\mathcal{A}$. Indeed, $X \times t$ is compact (the image of $X$ under $\mathbf{x} \mapsto (\mathbf{x}, t)$), so it is covered by finitely many $A_1, \ldots, A_k \in \mathcal{A}$. Their union $U$ is open and contains $X \times t$; by Theorem 4.6 some $\epsilon$-neighborhood of $X \times t$ lies in $U$, and in particular $X \times (t - \epsilon, t + \epsilon) \subset U$.

*Step 2.* By Step 1 choose for each $t$ an open interval $V_t \ni t$ such that $X \times V_t$ is covered by finitely many elements of $\mathcal{A}$. The $V_t$ cover $[a_n, b_n]$, so finitely many, $V_{t_1}, \ldots, V_{t_m}$, do. Then $Q$ is contained in $\bigcup_i X \times V_{t_i}$, each covered by finitely many elements of $\mathcal{A}$; so $Q$ is covered by finitely many. ■

**Theorem 4.9**

If $X$ is a closed and bounded subspace of $\mathbb{R}^n$, then $X$ is compact.

**Proof.** Let $\mathcal{A}$ be an open covering of $X$. Adjoin the open set $\mathbb{R}^n - X$ to get an open covering of $\mathbb{R}^n$. Since $X$ is bounded, it lies in a rectangle $Q$, which is compact (Lemma 4.8); finitely many members of the enlarged collection cover $Q$. Discarding $\mathbb{R}^n - X$ if it occurs leaves finitely many members of $\mathcal{A}$; they may not cover $Q$, but they cover $X$, since $\mathbb{R}^n - X$ contains no point of $X$. ■

All theorems of this section hold for arbitrary metric spaces in place of $\mathbb{R}^n, \mathbb{R}^m$, *except* Theorem 4.9.

## 4.2 Connected spaces

**Definition 4.4 (connected)**

A metric space $X$ is **connected** if $X$ cannot be written as the union of two disjoint non-empty sets $A$ and $B$, each of which is open in $X$.

**Theorem 4.10**

The closed interval $[a, b]$ of $\mathbb{R}$ is connected.

**Proof [Supplement].** Suppose $[a, b] = A \cup B$ with $A, B$ disjoint, non-empty and open in $[a, b]$ (hence also closed in $[a, b]$); say $a \in A$. Let $c = \sup\{x \in [a, b] : [a, x] \subset A\}$. Since $A$ is closed, $c \in A$. If $c < b$, openness of $A$ gives $[c, c + \delta) \subset A$ for some $\delta > 0$, so $[a, c + \delta/2] \subset A$, contradicting the definition of $c$. So $c = b$ and $[a, b] \subset A$, so $B = \emptyset$, a contradiction. ■

**Theorem 4.11 (Intermediate-value theorem)**

Let $X$ be connected. If $f : X \to Y$ is continuous, then $f(X)$ is a connected subspace of $Y$. In particular, if $\phi : X \to \mathbb{R}$ is continuous and $f(x_0) < r < f(x_1)$ for some points $x_0, x_1$ of $X$, then $f(x) = r$ for some point $x$ of $X$.

**Proof.** If $f(X) = A \cup B$ with $A, B$ disjoint and open in $f(X)$, then $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint, open in $X$, with union $X$; connectedness of $X$ forces one of them, hence one of $A, B$, to be empty. For $\phi$: let $A = \{y \in \mathbb{R} : y < r\}$, $B = \{y : y > r\}$, open in $\mathbb{R}$. If $r \notin \phi(X)$, then $\phi(X)$ is the union of the disjoint sets $\phi(X) \cap A$, $\phi(X) \cap B$, open in $\phi(X)$ and both non-empty, contradicting connectedness of $\phi(X)$. ■

**Definition 4.5 (line segment, convex)**

The **line segment** joining $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is the set of points $\mathbf{x} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$, $0 \le t \le 1$. A set $A \subset \mathbb{R}^n$ is **convex** if for every pair $\mathbf{a}, \mathbf{b} \in A$ the segment joining them lies in $A$.

A line segment is connected, being the image of $[0, 1]$ under $t \mapsto \mathbf{a} + t(\mathbf{b} - \mathbf{a})$. Every convex set $A$ is connected: if $A = U \cup V$ with $U, V$ disjoint, non-empty and open in $A$, pick $\mathbf{a} \in U$, $\mathbf{b} \in V$ and let $L$ be the segment joining them; then $U \cap L$ and $V \cap L$ are disjoint, non-empty, open in $L$, with union $L$, a contradiction. Hence open balls, open cubes and rectangles in $\mathbb{R}^n$ are connected.